



Metaphor as pedagogy in teacher education



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HIGHLIGHTS

- Metaphors illuminated students' shifts in understandings of curriculum.
- Metaphors informed instructors when students needed additional support.
- Metaphors allowed instructors to build relationships and honor holistic learning.
- Metaphors were individual, idiosyncratic, relevant, and supportive for students.
- The metaphors led to "quilted understandings" of curriculum.

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ABSTRACT

Two teacher educators used metaphors to create spaces for students to express their shifting understandings of curriculum over time. This paper centers upon their reflection and analysis of affordances and limitations of metaphor as a pedagogical strategy for supporting students. Instructors noted a) critical shifts in students' dispositions and understanding over time, b) meaningful insights into students' needs for additional support, c) how powerfully relationships were nurtured with students, d) and the quilted nature of the metaphors created, both across students and over time. Limitations are noted, as well as directions for future research.

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1. Introduction

As teacher educators working alongside of students, who are also novice teachers, in their urban classroom contexts during their vulnerable beginning years of teaching, we must support teachers as they strive to find answers to their own questions of practice and to develop themselves personally and professionally (Fisher, 2009). Novice teachers in the induction period have a range of complex learning tasks to engage in as they transition into classroom teachers (Fisher, 2009). This positions them in vulnerable circumstances as, simultaneously, they must develop networks of social, cognitive, and emotional support, find and negotiate resources available for their curriculum, understand and work to navigate expectations from administration and leadership (Feiman-Nemser, 2001; Flores & Day, 2006; Flores, 2006; Veenman, 1984; Zeichner & Gore, 1990).

Across the past six years, we have worked alongside Teach For America Corps Members who were learning to teach while they were provisionally-certified teachers in urban classrooms. Across two academic years, we have joined them both as instructional coaches in their classrooms as well as instructors among many of their graduate level courses, allowing us to know them more holistically as learners, teachers, and people. We have found it imperative that we create structures in our Master of Arts in Teaching program that intentionally support the cognitive, social, and emotional development and learning of students (Illeris, 2002). Simultaneously, it has proved important to provide engagements that encourage shifts in conceptual understandings and pedagogy and reflective practices over time.

As teacher educators working alongside students and their young learners in some of the most limiting and anti-democratically based schools in the United States, any tool for joining our students in imagining and building a different paradigm is one we are anxious to employ. We recognize that such an approach to supporting teachers through an interrogation in the metaphors they use related to teaching and learning is likely fruitful internationally as well. In our personal lives and our

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professional endeavors, we have found *the metaphor* to be a means through which we can come to more meaningfully, creatively, and hopefully, consider alternatives. Through this work of exploring the intentional use of metaphor as a pedagogical device with our students who are also novice teachers, we have come to see this as a space in our courses which opened expansive possibilities for local, national, and international education and teacher development.

As part of a several year inquiry into our teaching practices, we have considered the role and use of metaphors as spaces for students to reveal, to themselves and to others, their previously unquestioned positionalities towards their teaching and personal experiences. While in other work we use Critical Discourse Analysis (Fairclough, 2001; Janks, 1999) to analyze the content of the metaphors novice TFA teachers use as they think about curriculum (Fisher & Lynch, 2015), this manuscript delineates our own lessons as teacher educators, as we worked to understand the affordances and limitations of the use of metaphors as a snapshot of students' understandings which both revealed and at times obfuscated our understanding of the teachers we supported.

In many urban schools (including many of the schools where our students taught), the curriculum is handed to teachers in the form of a script or a specifically delineated pacing guide. In these contexts, our students need to have opportunities to think critically about their beliefs and understandings about learning, teaching, and curriculum in order to create innovative opportunities for relevant, meaningful, and responsive engagements with their students. As classroom teachers who interacted with curriculum and learners daily within these often limiting contexts, the students we worked with already had notions about curriculum, what it offered, how it might be constraining, and how it could be navigated. In the context of this class, students called those stances and perspectives by name, creating metaphors to articulate them, and shared these metaphors with us and with their colleagues in our program, building a complex and interwoven understanding and a multi-perspectival view of curriculum. Through this process, students articulated their own hopes for and struggles with agency and innovation. Elsewhere (Fisher & Lynch, 2015) we have discussed the findings from the content inquiry (See section 2.2 below), in which we analyzed the content of these metaphors of curriculum drafted by students and plotted each of the metaphors to consider and document shifts in teacher thinking about curriculum across time.

This manuscript, however, is a part of a simultaneous teacher-research inquiry (See section 2.3 below) we conducted. As we analyzed the content of the metaphors in the previous study, we engaged in consensus coding, creating an audit trail of the lessons and thinking that the use of metaphors made more and less possible for us as teacher-educators. This audit trail then became the data we analyzed for this inquiry. In this manuscript, we engaged in university-based teacher research (Cochran-Smith & Lytle, 1993) examining our own instruction and practice with the use of metaphors as a pedagogical tool for the support of teachers. We anticipate that it will contribute to the literature for teacher educators, making evident the complexities, limitations and affordances which accompany the use of metaphor as a pedagogical tool in support of teachers both within and beyond their initial certification. This means that we will not, in this manuscript, analyze the content of the metaphors or specific shifts made by participants (as we do in Fisher & Lynch, 2015), but rather we will focus on the findings about ourselves and our pedagogies which became evident through systematic inquiry into the pedagogical practice of metaphor use within a teacher-education program.

In this paper, we ask the question: What are the affordances and limitations experienced by teacher educators who employ metaphor as a pedagogical tool? Our purpose was to consider, from our

perspective as teacher educators, the ways in which metaphors facilitated some aspects of our instruction of a course centered on curriculum design while limiting others. We begin with a consideration of what metaphors do for and to us, as is suggested in the literature, including the work of international researchers and teacher development theorists. While there is a generous body of research related to the impact of metaphor on learners, we turn our gaze on the experiences we encountered when we carefully analyzed our own engagements as teacher educators employing this pedagogical practice, finding both hopeful and problematic aspects of this approach which was intended to offer space for teachers to critically (re)consider a ubiquitous aspect of each teacher's professional life and development.

1.1. *Metaphor in theory and research*

Metaphors hold promise for constructing and mediating understandings through analogy building (Steen, 2007). In other words, metaphors generate and support new meaning making by leveraging familiar, concrete reference points to develop nuanced understandings of novel, challenging, or esoteric constructs. Metaphor creation is innovative thought rooted in both the linguistic and conceptual forms. They are commonly treated as turns of phrases that serve primarily literary and aesthetic purposes. However, the Contemporary Theory of Metaphor (CTM), outlined by Lakoff (1993) suggests that metaphors are not merely linguistic expressions, but rather reflect the conceptual frameworks from which the speaker comes. They reveal the positionality from which we both form and express interpretations of concepts and experiences, regardless of one's native language, culture, nationality. Indeed, metaphors are not only used as linguistic flourishes, but "to invent, organize, and illuminate theoretical constructs" (Bowdle & Gentner, 2005). Fernandes (2008) posited that the main task of the metaphor is to create space for ideas, which are not easily wrapped around language accessible to the thinker/speaker, a construct particularly beneficial to inchoate and emerging ideas and beliefs. In effect, it seems that the very process of comparison and juxtaposition inherent in the creation or consideration of metaphor serves to make the unfamiliar familiar and to disrupt the overly familiar, causing it to become novel once again. In giving a construct a name that belongs to something else, metaphors offer coherence to our experiences by highlighting specific aspects of the nature of the ideas in contrast.

Educational researchers have recently interrogated the possibility of the use of metaphor to consider the thinking and development of pre-service and in-service teachers around the globe. Since metaphors simultaneously reveal, conceal, and reify often-unexamined and naïve stances, the metaphors created by teachers across their professional trajectories are increasingly considered in order to call awareness and often challenge teacher beliefs for the purposes of positioning teachers to align more intentionally their beliefs, language, and action. Such research has been conducted across cultural and linguistic contexts. Uysal, Burcak, Tepetas, and Akman (2014) analyzed the metaphors used by pre-service teachers within courses focused on developing strategies for classroom management. Ergin, Sahin, and Erisen (2013) considered the metaphors used by pre-service teachers in Turkey and found that these teacher candidates used metaphors describing their expectations of and experiences with children that positioned young learners as objects primed for molding and shaping by others, particularly the teachers themselves. Eren and Tekinarslan (2013) found that when they engaged in deconstruction of metaphors used for learning and teaching within the teacher preparation coursework they offered, teacher candidates considered the cognitive and affective implications of the metaphors they

chose. We have joined several international researchers, such as those above, in wondering what intentional consideration of the language used in metaphors for teaching, learning, and learners might reveal about the stances, dispositions, positions, and developing understandings of students (Fisher & Lynch, 2015). This current inquiry varies from other previous research on teachers (both preservice and in-service) content and use of metaphors, as it turns the gaze toward the findings we encountered as we considered the implications of the use of metaphors as a pedagogical engagement and the ways we, as teacher educators, experienced and shifted through the use of this practice.

Let us consider an example one of us recently read in an assignment written by a teacher just entering a doctoral program. The teacher wrote: “I wanted to be a teacher so that I can shape the minds of future generations.” The metaphor here presents a mind as something to be molded. Considering this from one perspective, we might understand this teacher as viewing learning as a process of transformation; we are not the same after having gained access to new ideas, strategies, and understandings, much like clay takes on a new shape in the hands of a sculptor. From another perspective, one that challenges and calls into question assumptions revealed in metaphor, we might wonder about the role of agency in learning. Who is doing the shaping? Who determines whether we sculpt a vase or a brick? Who benefits from the new form? We might find it troubling that learners here are positioned as objects to be acted upon. Were we to talk with this teacher about the questions we pose, she might be shocked to hear such dispositions expressed (or interpreted by others) due to her language choices. This disorientation might lead her to reevaluate underlying assumptions she held related to the importance of agency in learning or might cause her to rethink the language she uses to describe the process.

From this example, we demonstrate the ways that a close consideration of metaphor can shed light on the underlying, and possibly least questioned, beliefs and understandings expressed by the authors of metaphors. In addition, a careful consideration of the metaphors we use to articulate experiences and views allow us to reveal to ourselves aspects of our thought that may have been previously unquestioned. We contend, along with Lakoff (1993) that as we appropriate metaphors into our discourses and expressions, we unwittingly adopt conceptual frameworks they express, either implicitly or explicitly. Put another way, as metaphors shape our words, they, in turn, shape our understandings of meaning-making within our world.

Metaphors are constantly shaping our thoughts and actions in ways that are simultaneously creative and restrictive (Haggis, 2004), opening new ways of thinking and simultaneously preventing us from considering or attending to aspects of the construct under consideration that are not consistent with the chosen metaphor (Lakoff & Johnson, 1980).

1.2. Metaphors and critical pedagogy

As critical pedagogues, we nod our heads with Paulo Freire (2000/1972) as he has posited the importance of developing strategies of reading both the word and the world from a perspective that names and investigates implications for power, powerlessness, and consciousness-raising. Understanding that language and thought have mutually informative relationships (Vygotsky, 1986/1934) that impact action, we have found metaphors to offer powerful intersections between the three. By challenging ourselves to deeply investigate the language choices within out metaphors and how they do or do not align with what we believe about the world around us, we are better positioned to critically read our words and worlds. Indeed, because of the boundedness of

metaphors (e.g. a turn of phrase, rather than larger language practices), they become a microcosm within which we come to wonder about how we are living into the ideals of critical pedagogy that we, the researchers, and the program in which we teach, claim to value. This is especially valuable when discussing with novice teachers ideas so largely discussed in public forums, but so rarely critically challenged, as are many ideas related to schooling. For these reasons, we wanted to investigate the roles of metaphor in our work alongside our students, all of who were novice teachers both finishing their initial certification and Master's degree while completing their early years of elementary school teaching.

2. Methods: process and pedagogy

Forty-six students were enrolled in the curriculum course, with thirty-two of those providing IRB consent for the use of their data for research purposes, and twenty-seven of those evidencing their thinking across time by following the general directions of the engagement. While we certainly learned a great deal about our students through the process which helped us support them in their future practice and coursework, we did not deconstruct through formal analysis either the content of their metaphors or our lessons from this engagement until all students had completed coursework at our university.

We have utilized metaphors for over seven years in the context of our classes, often as a synthesizing ‘ticket out the door’ or to initiate discussions within small groups unpacking a variety of experiences and understandings. However, we had not systematically considered the implications of this practice for our work as teacher educators endeavoring to support teacher development. This inquiry into our own lessons and deconstruction of the practice from our perspective as teacher educators is the focus of this manuscript.

2.1. Context

The specific metaphors we systematically considered were drawn from a course assignment that was part of a teacher-preparation program at a large university in the Southeast United States. We, the researchers, co-taught a six credit hour course, *Integrative and Iterative Curriculum Design*, across a four-month period. It is of note that we created and designed this course for 46 novice teachers who were simultaneously graduate students and Teach For America Corps Members, serving as novice teachers in elementary schools in our community. Due to their multiple identities (as novice teachers and as our students in certification/masters coursework) they are referred to using both of these identifiers throughout the manuscript.

Within this class, our students met in a book club three times to discuss Wiggins and McTighe's (2004) *Understanding by Design*, met with faculty ten times for focused engagements around literacy in the content area and Science and Social Studies methods. Additionally, they met for three full days with faculty for targeted conversations around curriculum design and responsive planning. After each book club or curriculum focused class, students were asked to create a metaphor for their current understandings of curriculum. The example initially provided for them during the first day of the course was, “Curriculum is an onion with so many layers to peel back.”

As we crafted the course, we decided to have each student submit his or her metaphor on padlet.com before each class, acting as an online bulletin board where students both posted their own metaphors and viewed the metaphors and connections their classmates made (See an example in Fig. 1.). Padlet allows users to post a very small number of characters, so students carefully crafted

their language for this space and a shared audience.

We did not grade metaphors individually or for quality. Rather, students compiled each of their metaphors and submitted them together at the end of the course, earning full credit for engaging in the thinking and production of metaphors across time. This provided students with a sense of freedom and a space for authenticity which might have been compromised if they thought that the content of their metaphors might negatively impact their grades. These periodic metaphors provided springboards for class discussions and reflections. As instructors, we noted how this assignment, while not extensive, offered insights into the thinking and developing understandings of our students. As we worked alongside these same students again the following two semesters in coursework and in their classrooms, it was evident from the pedagogical changes many of them had made and from their discussions and explications of those changes that many students had shifted significantly in their thinking about curriculum, becoming more centered on student learning and inquiries. Upon the graduation of this cohort, we decided to return to this assignment to consider the engagement of metaphor creation as a pedagogical practice.

2.2. Initial inquiry

In a previous inquiry (Fisher & Lynch, 2015), we employed

Critical Discourse Analysis (Fairclough, 2001; Janks, 1999) to analyze the shifting discursive moves novice teachers made as they constructed and shared metaphors about curriculum, agency, and learning. Through the hermeneutic process of consensus coding (Hill, Knox, Thompson, & Hess, 2005), we discovered trends across the developmental shifts in understandings of the novice teachers we supported. Analysis from the previous inquiry (Fisher & Lynch, 2015) indicated that within the context of our curriculum class, students demonstrated interesting shifts in their thought and dispositions towards curriculum design and development across the semester. As we more fully outline elsewhere (Fisher & Lynch, 2015), it became evident in our analysis that the most meaningful shifts could best be described as falling along two continua. We refer to one continuum as *agency*, since almost each metaphor that the 27 students in this study composed revealed assumptions of agency as residing within the teacher, external to the classroom, or a balanced combination of both. Similarly, a second continuum, referred to as *pliancy*, allowed students to explore understandings of curriculum as a static or flexible concept or possible both.

As an example of how these concepts appeared in students' metaphors, we might consider Tabitha's initial metaphor (all names are pseudonyms): "Curriculum is like a musical composition that integrates the sound of so many instruments and the people that play them." From our perspective and based on what we know of



Fig. 1. In response to reflection on readings and discussions of "curriculum," students posted original metaphors and similes to Padlet.com.

Tabitha, we see a disposition reflected here on a measure of flexibility in curriculum; rather than seeing curriculum as a lock-step process from which she cannot deviate, Tabitha seems to view it as something that must take into consideration the instruments and people involved along the way. In addition, we interpreted her use of the word “composition” to recognize as a combined effort of a composer, the one who provides the foundation of a shared score of music for which the musicians to perform, and the musicians, who are to take the musical score and perform it as their own. From this, we interpret Tabitha’s metaphor to see curricula as externally structured but performed interpretively by the teacher. We use this example to demonstrate the ways in which we analyzed all participating students’ metaphors. In exploring the pliancy and agency of their metaphors in the previous inquiry (Fisher & Lynch, 2015), we found important shifts occurring across the semester which led us to questions about the use of this practice (of the creation of metaphors with students who are also novice teachers) more broadly.

While we have, elsewhere (Fisher & Lynch, 2015), considered the construct of curriculum as understood by these novice teachers, we found it important that the use of metaphors pedagogically seemed to have implications as an engagement with teachers who were grappling with many complex constructs of teaching and learning which might benefit from an opportunity to deconstruct, both linguistically and holistically. This question led to this second inquiry, described within this manuscript, wherein we examine the use of metaphor as a pedagogical practice seemed to do to and for us, and for the students we supported.

2.3. Current inquiry

This inquiry is grounded in teacher research paradigms, as we, the instructors of this course, considered and investigated our own teaching practices. Cochran-Smith and Lytle (1993) encourage educators of all levels to engage in research that is conducted in the classroom contexts by those engaging with learners directly so as to investigate a phenomenon or problem they notice and to strengthen their own practice. As we began to note important shifts among students’ language and understandings, we became committed to better understanding what this shift meant in regards to our instruction and their learning across the semester. This university-based teacher research created opportunities for critical reflection, offering us an opportunity to look carefully and intentionally at our instruction, considering ways metaphors evidenced learning and supported students, and analyzing language use and shifts in thinking and practice both within and beyond the context of this course. As teacher educators, we view this type of analysis as both a moral and vocational imperative as it offers us a critical opportunity to develop ourselves and our practice on behalf of the novice teachers we support and the P-12 learners they serve.

As is often the case, as co-teachers and co-researchers, while we engaged in the consensus coding process (Hill et al., 2005) of the first inquiry, we simultaneously found ourselves pulled into conversations about what—and who—this pedagogical practice seemed to make more visible and less visible. Through these collaborative coding conversations and the audit trail that we created to record these coding conversations, it became clear to us that the findings of this inquiry included more than a consideration of the continuum from which each novice teacher spoke about curriculum. As we read each metaphor and discussed it, we wrote our open analytical codes in an excel document, noting in one column our interpretation of the metaphor and in another the lessons that we gained thinking about the metaphor as a tool to come to know and make our students more visible. In effect, through documentation of our consensual coding process and discussions, we crafted an

audit trail of the pedagogical findings revealed through each metaphor. This audit trail then became the data we utilized for the current inquiry into our own practice.

After working through each of the metaphors from the data set, open coding the discoveries about our pedagogy that were revealed within individual metaphors and across the five metaphors drafted by each individual, we reread our codes related to what the metaphors revealed about the use of this pedagogical approach. Together we clarified these open codes into categories and themes and evaluated each theme for salience and resilience across the data set (Lincoln & Guba, 1985; Strauss & Corbin, 1998).

This university-based teacher research became focused on how we ourselves were coming to understand the use of metaphor in learning and teaching. Investigating students’ growth and cognitive tensions allowed us to grapple with and seek patterns in our own practice and thinking. As a result, we were led to consider the role of metaphor as a pedagogical tool for coming to know, understand, and support our students, including a consideration of the affordances and limitations of using metaphor as a pedagogical strategy in teacher education courses and teacher development opportunities.

3. Results: metaphor as pedagogical tool

By analyzing the data of our university-based teacher inquiry, we found that offering students opportunities to reflect through the use of metaphor offered us (as faculty) and them (as students and as novice teachers) particular affordances. Themes that arose indicated that students’ metaphors offered us, as instructors (a) an audit trail each students’ engagement with ideas over time; (b) insights into students who needed additional support with course content; and (c) enhanced and more holistic knowledge of individual students. Metaphors provided students with opportunities to express their learning in idiosyncratic, aesthetic, personally relevant, and meaningful ways. However, it is important to note that we encountered limitations to this pedagogical approach when we had limited familiarity with individual students.

3.1. Metaphors illuminated students’ engagements with ideas across time

Teacher development and shifts in beliefs and understandings are non-linear, incremental, and occurs across time (Fisher & Lynch, 2015). Often these shifts are difficult to note and document, as there are rarely definitive points that illuminate positions and stances as they gradually shift (or remain the same). These metaphors, which were crafted across a four-month period, served as crumbs marking and revealing the journey taken by individual students as they articulated their beliefs and attitudes about curriculum across time. It is important to note that students were not asked to engage in the crafting of these metaphors daily or weekly. Rather, they were situated periodically at critical moments during the class when they had read or discussed a set of ideas which might inspire a new or varying perspective on the purpose and design of curriculum. We found that these metaphors served to articulate both the process and the content of the shift.

David provided us with a sequence of metaphors that gave us helpful insights into the shifts in understanding that his particular journey through the course content took. At the time, David was between his first and second years in the elementary classroom where he supported English language learners across grade levels. We have compiled his metaphors from across the semester in Table 1.

Upon considering the nature of David’s metaphors and how they comment on the nature of *curriculum*, we noted the rigid nature of

his first two metaphors. If curriculum is like a Transformer or a puzzle, we considered the nature of these two items: they are pre-constructed. They are made to be manipulated by the player, yet they are intended to be handled in very particular ways, as well as to challenge him or her. Transformers are combat-centered toys; puzzles are designed to challenge one's ability to analyze and locate patterns. While both are toys, David's description of them focuses on a mental and physical trial and confusion as a possibility, depending on one's ability to navigate them. In both cases, curriculum seemed to represent challenge, puzzlement, and a pre-determined object given to a user, as is often the case in script-centered instructional initiatives and step-lock scope-and-sequence plans offered to many novice teachers.

In the two subsequent metaphors, after reading about curriculum and discussing it in his book club and in our course sessions, we see David beginning to create metaphors which imply that at these points in his thinking, he conceptualized curriculum as requiring an element of flexibility. The food metaphors associated curriculum with nurturance, pleasure (focusing on cake and a "delicacy"), and processes. This is a striking contrast to the previous toy metaphors that emphasized a stagnant product orientation with curriculum. Instead, David describes curriculum as offering the user with options, control, and an enjoyable path to getting there. The cake metaphor refers to a measure of calculation, but acknowledges room for spontaneity, as one might do with a recipe he or she wants to experiment with in new ways. The image of tasting along the way in the fourth metaphor is the first time David has alluded to a measure of responsiveness in the development of curriculum. Once he sees curriculum as both product *and content*, he is able to acknowledge that there can be freedom and enjoyment interwoven into the more procedural ingredients of what he teaches.

The final metaphor represents a more complex understanding of curriculum than previously expressed by David. In the presence of a pre-designed Swiss army knife, we understand David to view curriculum as structured by external forces; since the knife transforms in response to the user's situated needs, he expresses a simultaneous measure of authority and ownership. There is both rigidity and flexibility as the user works with the item as it was given to him, yet has great creativity with how he uses the objects. For example, one might just as easily use the corkscrew tool for self-defense or to clean under fingernails as to open a bottle of wine. This recognition of flexibility and ownership within a potentially rigid structure represents David's ability to view the many purposes, perspectives, applications, and understandings of curriculum, a key goal of this particular course.

This is merely one example of the kinds of shifts we saw among students. While many joined David in his developing complexities regarding curriculum, others created sequences of metaphors that emphasized only one aspect over the other possibilities. When this was the case, we tended to have enough evidence through the audit trail provided across metaphors to identify where teachers perceived their notions of curriculum to be limited.

3.2. *Metaphors let us know when students needed additional support*

While this assignment was one that we asked students to consider across time, thinking about the construct of curriculum as they engaged in their daily lives, we also understood that this was not a traditional assignment that required a lengthy essay or a substantial commitment of time. At several points across the semester, it seemed evident to us that some novice teachers were not engaging in this endeavor with a sense of seriousness as evidenced through unexamined or unreflective metaphors. Emma's second metaphor was "Curriculum is like a water bottle—it can be too hot, too cold, or just right." This metaphor, and others like it, demonstrated a simplistic understanding of curriculum and told us very little about what Emma understood and was coming to understand. Amy, in her third metaphor, said "Curriculum is like the lettuce in a salad. You need it to call it a salad but it would be one bland salad without other ingredients!" We grappled with this metaphor in that it indicated that Amy had not thought deeply about the referent, since lettuce is not inherently required in all types of salad, and we could not actually determine what this metaphor indicated about Amy's thoughts about and experiences of curriculum.

In these cases, a lack of care or understanding of the referent point made for a confusing or minimal demonstration of the evolving thinking of the candidate. Occasionally, novice teachers would merely define curriculum rather than likening it to another thing. Twice across the course, Susan defined curriculum saying "[Curriculum is] ... a focus on the information novice teachers need to be able to transfer their skills" in her second metaphor, and "[Curriculum is] ... reflection in action" in her fourth metaphor. It must be noted, however, that no candidate consistently evidenced this type of minimal consideration across time. These were unusual events, though, and indicated to us in retrospect that we likely should have ensured that all novice teachers fully understood what we were asking for.

Sometimes, metaphors illuminated more troubling underlying assumptions about learning and teaching. At one point, one of the novice teachers posited that "Curriculum design is like farming; much of the work is done upfront to reap a harvest in the end (student achievement)." We are particularly troubled by the concept of "harvesting student achievement" and the ways in which that phrase simultaneously objectifies students and owns their success. This metaphor positions teacher as the agent and the students as the objects. Rather than engaging in the work and benefiting in the harvest, the students themselves are positioned as the cash crop or harvest.

We, the instructors, found the insights or misapprehensions reflected in such metaphors as fruitful tools in the coaching we conducted at students' sides when we rejoined them in their classrooms in the fall following this course. While we did not coach every student in the course, between the two of us, we coached the majority of them, joining them monthly in their classrooms, where we supported them in goal setting, narrative feedback, and engaging in increasingly responsive and research-based

Table 1
David's compiled metaphors.

Number	Metaphor
1	Curriculum is like a Transformer: it can be your best ally that makes your life easier (Autobot) or your worst enemy that complicates everything (Decepticon).
2	Curriculum is like a puzzle that can seem confusing when working with all of the pieces individually but that will hopefully look nice when complete.
3	Curriculum is like baking a cake. You know what you want the end result to be, but there are certain calculated (and sometimes not-so-calculated) steps to get there.
4	Curriculum is like cooking—there are recommended steps and ingredients that you should include, but you should constantly taste what you're making in order to make sure that your delicacy is how you want it to be.
5	Curriculum is like a Swiss army knife that affords you different tools and weapons, depending on what is most pressing at the moment in your world.

instruction. In addition, we read reflections our students wrote daily about their experiences in their classrooms. A key affordance of our use of metaphor across the curriculum course was that we had greater awareness of students' language use both in their teaching contexts and in their writing about them. We were able to help students unpack metaphors we noticed in ways that helped them reveal to themselves and to us the beliefs undergirding them. We were also able to harken back to ideas they grappled with in the curriculum course since their metaphors often provided us with condensed representations of the kinds of teachers they saw themselves to be or strive to be. In short, we got efficient mileage out of our students' metaphors, both immediately in the semester and across our future support of them across another academic year.

3.3. Metaphors helped us know students better as individuals

At times, students shared with us metaphors which offered us windows into their interests, priorities, life stories, beliefs, and the ideas and constructs which occupied their thinking (at least at the moment of the creation of the metaphor). Through these metaphors, we came to know novice teachers better as we developed a more nuanced perspective about them as full individuals with rich lives outside of our shared context. As teacher educators who are about a decade older than most of these novice teachers, we realized, at times, that we did not share the same frames of reference as they have. One of the most nearly incomprehensible metaphors was so because we did not have a great deal of knowledge about the interests, experiences, and history of some of these novice teachers. David, whose metaphors were considered across time above, made this very obvious to us as he stated in his first metaphor, "Curriculum is like a Transformer: it can be your best ally that makes your life easier (Autobot) or your worst enemy that complicates everything (Decepticon)." Terry, his coach and instructor, had truly no idea about the referents (Autobot and Decepticon?) and was reminded that she had much to learn from and about her students, their lives, interests, and histories.

In Holly's first metaphor, she said "Curriculum should be like an orange ... All of the pieces of content should fit together to create one delicious product!" Her use of the word "should" indicated to us that she was grappling with a more rigid understanding of how curriculum could unfold and that, in our continued collaboration, we should continue to support her as she envisions a more open and hopeful position toward a range of possibilities. Conversely, in Shannon's fourth metaphor, "Curriculum is freedom OR constraint. How you create it, follow it, or diverge from it may either set your learners free with flexibility, or enslave them within a strict plan," we discovered the extent to which *Pedagogy of the Oppressed* (Freire, 2000/1972), a reading we engaged in with these particular novice teachers, was shaping her thinking and her teaching, a reality we might not have been privy to had she not shared this connection through her metaphor.

3.4. Metaphors created opportunities for aesthetic and personalized understandings

Aristotle described the complex make-up of metaphors which combine "clarity and sweetness and strangeness and its use cannot be learned from anyone else." (The Rhetoric, III, trans. by George A. Kennedy, 1991). The metaphors each individual shared reflected her unique understandings of her world and therefore was a unique manifestation of divergent thinking connected to her lived experiences and developing understandings. In addition, the more symmetrical the metaphor seemed to be, meaning the analogy between the objects being compared was tightly wound (Bowdle &

Gentner, 2005), the more artful and aesthetic it seemed to be. The merging of "clarity and sweetness and strangeness" offered novice teachers opportunities for self-revelation, creativity, and relevance in interesting ways. As both an intellectual and affective construction, we found that students' metaphors often depicted understandings aesthetically.

Jasmine offers an example of a metaphor we found particularly artful. In her second metaphor, she stated "Curriculum is like a garden of wildflowers. If watered and cultivated, the different flowers complement each other and provide for the many insects in the garden. If over-planned and micromanaged, the garden appears too controlled and may not meet the needs of the insects and animals in the area. If curriculum is too tightly controlled by the teacher, the students will not be able to have input and truly flourish in their cognitive development." Jasmine's metaphors were nearly always marked with an interpretive piece, in which she helped the reader follow her comparison to teaching and learning and curriculum specifically by explaining the cognitive connection behind her aesthetic metaphor explicitly. This specifically served as a lesson to us as we considered (and will discuss below in section 3.5) limitations we encountered through the use of this pedagogical approach.

Through our analysis, it seemed evident that the creation of metaphors as a course engagement offered space for these novice teachers to enter the discussion as holistic individuals, reflecting upon their positions and ideas in uniquely and inherently personal ways which represented both their head and their heart. This individual association offered many spaces for them to tap into their funds of knowledge (Moll, Amanti, Neff, & González, 1992), share their interests, and offer us perspectives on what they valued and windows into their lives, while creating a unique and idiosyncratic representation of their understandings. This transaction was unique not only to the individual, but also to the individual at a specific time and space, shifting in interpretation even across one course as their conceptualization shifted.

3.5. Limitation: less familiarity with some students led to imposed interpretations

When engaging in this type of inquiry, it is imperative to consider the ethical considerations of this type of analysis of the content and meaning of metaphors and to acknowledge the limitations inherent in this approach. Throughout our analysis, we were reminded of Reader Response (Rosenblatt, 1978) theorists' understanding of the meaning of text residing not merely in the printed word, not only in the reader's mind, but in the transaction, or interplay, between the two. Similarly, our reading of novice teachers' metaphors required us to approach the text based on our knowledge and contexts in relation to the author. We realized that not only was each student's creation of the metaphor an instance of Rosenblatt's construct of transaction (1978), but also that without clarification and connection/explanation of the metaphor on the part of the student, our interpretation of those metaphors (and the authors thereof) was a transaction as well.

To add additional complexity in considering the limitations and affordances of the metaphor as a pedagogical tool, we noted that as we engaged in the analysis, we were able to bring to bear our previous knowledge of some of the students more than others. There are many whom we had taught in previous coursework, coached in their classroom context, and engaged in frequent conversations about teaching and learning. Frankly, we knew them more deeply on a personal level than students with whom we had less frequent contact. Jennifer, for example, noted that "curriculum is like a fence." Both of us know Jennifer well, having worked closely with her across courses, and contexts. Therefore, we knew

with clarity that near the time of the writing of this metaphor, Jennifer had recently scaled and jumped a fence, seriously breaking her leg. Acknowledging that Jennifer seemed not to feel fenced in and controlled by this physical barrier, but that she leaped, in fact to her peril over to the other side, made for a more nuanced and informed, though potentially less transparent reading of this metaphor. Knowing many of these particular students very well, we cannot, and likely should not, pretend that we do not imbibe the reading and interpretation of student's metaphors with our knowledge of who they are and what we know of their story.

That being said, we also realized that at times we did not have enough understanding of the authors' relationship with the referent to determine the feelings and beliefs they were implying through their metaphor. The clearest example of this category was Janet's third metaphor, in which she stated "Curriculum is like love. You have to experience it to understand its design." Certainly, we have very little knowledge of Janet's experience with her chosen reference point. It does seem, however, that this metaphor presumes *one* experience, *one* design, and the uniformity of love. Clearly, we cannot assume that we, the readers of Janet's metaphor, share Janet's meaning of, experience with, or understanding of love. At times, our knowledge of the range of experiences and multiplicity of stories of these particular novice teachers which we did not share made it very difficult to interpret the intent of their metaphors. As theorized by Rosenblatt (1978), we help create the poem when we reinterpret what the authors have encoded in their metaphors.

For students we knew less well, we found ourselves during the analysis of their metaphors, in sometimes ethically troubling positions of realizing that we were imposing our own meanings upon their words. While we feel confident that, at each other's sides, we met each metaphor with positive intentions and dispositions, students did not have opportunities to *talk back* to our interpretations. The same was true during our instruction, as well, particularly for those students who, because of our limited familiarity with them as individuals, might have been less willing to correct our possible misunderstandings of their ideas and words. This limitation has guided our thinking as we consider future instructional approaches with and uses of metaphors.

One specific modification in our practices which was a direct shift based on this limitation and the lessons learned during this process was the need to have the authors of the metaphors themselves compile, consider, and reflect upon the meaning of the collection across time. Considering what students meant by individual metaphors was helpful, but we had not, before the crafting of this article, asked the authors of the metaphors to speak to the corpus of their considerations and what implications they found as they reflected on their stance about the construct.

Since this inquiry, this pulling together the tethers of metaphors across time and having the author consider what can be understood about their understandings has been a critical aspect of this engagement. In the years since the collection of this data, we have added a component of this engagement where students compiled and wrote a short reflection based on their multiple metaphors, describing what they noticed about their evolving thinking across time and patterns in their thinking that were evidenced in the collection of their metaphors. This additional opportunity for them to consider and analyze their own thinking as revealed in these short metaphors provided space for consideration of their growth and growing edges across time, not only for us as their professors, but also for students themselves.

To be sure, we have been asked by students why we request that they draft multiple metaphors related to one topic. We feel that asking students to draft multiple metaphors across the class not only revealed the complexity of teaching and learning, but it was

more meaningful than if the students had only drafted one metaphor. A single metaphor would only reflect the momentary position and stance that was held within the confluence of their thinking and their living in the world. Having several opportunities to draft metaphors around a single topic provided a more complex and in-motion vantage for consideration of their thinking over time and with increased experience and deconstruction of this aspect of schooling. Our more recent practice of providing them with a time to look across their own thinking and consider their stances through the short reflection created an opportunity for students to see and take more meaning from the process.

3.6. Pulling the themes together: quilted understandings of metaphor-as-pedagogy

"Curriculum is like a quilt that is made up of many carefully crafted and precisely measured pieces of fabric, forming one unified blanket." These words were composed by one of the students we supported in her teacher-preparation program. In reading them, we found them to offer a compelling way to express and understand not only curriculum, but our experience in harnessing metaphors to support professional reflection and growth within our courses. It is perhaps not surprising that, in making sense of the lessons and learning we can experience from metaphors, we found ourselves turning to a metaphor to make evident and explicate the complexity of our developing understandings.

When creating a quilt, the artisan must return to pre-existing scraps of cloth, re-imagining them at the side of other patches and intentionally deciding how to best pair these patches together, so as to emphasize particular aspects of each in complementary patterns. By the end of the stitching process, the artisan has created something new from something old. She has created a piece of art that is as useful in a cold winter as it is aesthetic on a summer afternoon folded on the back of a chair. She engages a sense of order and control in creating the most useful quilt that she can, as well as a sense of artfulness and craft. In analyzing our memos drafted through our use of CDA in considering the content of students' metaphors and what the use of this pedagogical tool offered both us (as instructors) and them (as students of teaching and teachers of students), we believe their metaphor crafting provided an intentional engagement which allowed them opportunities to return to, revise, and re-purpose their ideas and to merge the head and the heart. We consider this type of idiosyncratic representation of understanding (evolving over time) as a possible learning opportunity for both cognitively and personally meaningful insights to be fostered, developed, and shared.

This quilting process or repurposing and reconsidering previous thinking and beliefs seemed to be part of each student's individual process. Upon planning the curriculum course, we had taken intentional steps to relate the content of this course to prior experiences we had provided students in previous courses. We found them metaphorically *stitching* together ideas from their professional reflections across their first year in the classroom, returning back to lessons they'd taught, and building upon ideas from courses they had taken on literacy, critical pedagogy, etc. Furthermore, as quilters tend to do, they often engaged in the tedious, but transformative, work of tearing out stitches that no longer seemed to fit the pattern. By this, we mean that we often saw students using metaphors to begin to separate themselves from ideas they began to find tensions within (e.g. "Curricula is a prescription ..."). Indeed, the process of pulling away from ideas and theories that no longer fit their shifting and increasingly complex understandings was some of the most meaningful work we witnessed from students.

In addition to the internal quilting of ideas together, a measure of quilting began to emerge as students posted to the Padlet board.

They logged on and wrote their metaphor on a virtual notecard that was pinned to the board for their classmates to see. As post after post was displayed, these square patches not only visually reminded us of a quilt, but their content as well simulated a verbal quilt (see Fig. 1). Ideas representing different dispositions and experiences lay side-by-side, very much representing the constructivist framework that guides so much of our teaching and learning experiences as the course instructors.

As the course progressed, we presented students with content that led to shared discourses and bodies of knowledge, about which they discussed, reflected, and wrote regularly during the semester. We might have expected (or required) that our quilt to converge into a synthesized tapestry. However, because our students were not graded based on the content of their metaphors, they seemed to feel free to authentically represent their thinking and stance at each particular juncture. This appeared to be evident through the range of metaphors crafted and stances evidenced. Just as constructivism and transactional theory posit, due to the diversity of experiences each student brought to her learning and the idiosyncratic interpretations and transactions between her own lived experiences and course content, the originality remained, each patch a bold representation of our students' stances—boldly unique patterns, shades, and textures.

4. Discussion for teacher educators

We have found that both our experiences and analysis of data pertaining to metaphors as a pedagogical tool offered opportunities for students to explore revealing and critical relationships between their experiences, their understandings, and the language they used to express and investigate those stances. Moreover, the use of these metaphors as a pedagogical tool in teacher education courses offered us, as instructors, opportunities to blur the line between relational and content-centered understandings of our students along the way. Indeed, there is room for future considerations regarding teacher educators' use of metaphors in support of students. While metaphor has been considered in teacher development with respect to learners' experiences across context and cultures (e.g. Eren & Tekinarslan, 2013; Ergin, Sahin, & Erisen, 2013), we feel it is important that, as a field, we consider our own goals, dispositions, and directions as we engage alongside of students in the consideration of metaphors.

Teacher educators engaging in the tradition of teacher research (Cochran-Smith & Lytle, 1993) might find the following possibilities fruitful as they consider the meaning and challenges of the use of metaphor as a learning and teaching tool in their own practice alongside preservice and in-service teachers. It might be a fruitful investigation to offer benchmark opportunities for teacher candidates to craft, analyze, expand, revise, and discuss (either with faculty or classmates) metaphors representing their understandings of teaching and learning. In the context of individual teacher education courses, instructors might investigate their own use of guiding metaphors to prepare students for discussions of readings and teaching experiences. They might weave such guiding metaphors into portfolios, projects, or other representations of knowledge. As a strategy for considering educational theory and policy, teacher educators might draw students' attention to metaphors common within particular conversations (e.g. metaphors as they exist in discussions of the standards movement or within the grouping of learners) and directly expand the various layers of the comparison with the intention of critically evaluating both language and content. When evaluating writing, we might call the writer's attention to metaphors that contradict stated or enacted beliefs for the purpose of revision of either selected words or

beliefs. By engaging in intentional inquiry into the use of metaphors as a pedagogical tool in the support and development of teachers and the ways that meaning is constructed, shared, and reconsidered through this pedagogy we might add to the growing patchwork of our own understandings not only of what metaphors do for and to us, but also of ourselves as educators and teacher development practices as a field.

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